Name: K. Archana **Reg No:** 192524209
Course: Computer Vision **Code: DSA0204**

CO4 AT3

1. Object Detection Performance – Viola-Jones vs YOLO
Given Data:

Model A (Viola-Jones): TP = 80, FP = 20, FN = 30

Model B (YOLO): TP = 90, FP = 15, FN = 20

(a) Calculate Precision and Recall

Precision Formula:

Precision = TP / (TP + FP) × 100

Viola-Jones Precision:

= 80 / (80 + 20) × 100

= 80 / 100 × 100

= 80%

Recall Formula:

Recall = TP / (TP + FN) × 100

Viola-Jones Recall:

= 80 / (80 + 30) × 100

= 80 / 110 × 100

= 72.73%

YOLO Precision:

= 90 / (90 + 15) × 100

= 90 / 105 × 100

= 85.71%

YOLO Recall:

= 90 / (90 + 20) × 100

= 90 / 110 × 100

= 81.82%

Comparison:

Viola-Jones: Precision = 80%, Recall = 72.73%

YOLO: Precision = 85.71%, Recall = 81.82%

Conclusion:

YOLO is better because it has both higher precision and higher recall.

2. Motion Estimation Error Analysis

Given Data:

Frame 1: Method A = 2 pixels, Method B = 3 pixels

Frame 2: Method A = 3 pixels, Method B = 5 pixels

Frame 3: Method A = 2 pixels, Method B = 4 pixels

(a) Calculate Average Error

Average Error Formula:

Average Error = Sum of Errors / Number of Frames

Method A:

= (2 + 3 + 2) / 3

= 7 / 3

= 2.33 pixels

Method B:

= (3 + 5 + 4) / 3

= 12 / 3

= 4.00 pixels

Comparison:

Method A = 2.33 pixels

Method B = 4.00 pixels

Conclusion:

Method A is more accurate because it has a lower motion estimation error.

3. Optical Flow Magnitude Comparison

Given Data:

R1: Method A = 5, Method B = 8 pixels/frame

R2: Method A = 6, Method B = 10 pixels/frame

R3: Method A = 5, Method B = 9 pixels/frame

(a) Calculate Average Motion Magnitude

Average Motion Magnitude Formula:

Average Magnitude = Sum of Motion Magnitudes / Number of Regions

Method A:

= (5 + 6 + 5) / 3

= 16 / 3

= 5.33 pixels/frame

Method B:

= (8 + 10 + 9) / 3

= 27 / 3

= 9 pixels/frame

Comparison:

Method A = 5.33 pixels/frame

Method B = 9 pixels/frame

Conclusion:

Method B captures stronger apparent motion because its average optical-flow magnitude is higher.

Note:

Higher magnitude means stronger measured motion, not necessarily higher accuracy.

4. Depth Estimation Comparison – Stereo Vision vs Structure from Motion

Given Data:

Object A: Actual = 2.0 m, Stereo = 2.0 m, SfM = 2.5 m

Object B: Actual = 3.0 m, Stereo = 3.0 m, SfM = 3.8 m

Object C: Actual = 4.0 m, Stereo = 4.0 m, SfM = 5.0 m

(a) Calculate Absolute Error

Absolute Error Formula:

Absolute Error = |Estimated Depth − Actual Depth|

Stereo Vision – Object A:

= |2.0 − 2.0|

$= 0$ m

Object B:

$= |3.0 - 3.0|$

$= 0$ m

Object C:

$= |4.0 - 4.0|$

$= 0$ m

Average Stereo Error:

$= (0 + 0 + 0) / 3$

$= 0$ m

SfM – Object A:

$= |2.5 - 2.0|$

$= 0.5$ m

Object B:

$= |3.8 - 3.0|$

$= 0.8$ m

Object C:

= |5.0 − 4.0|

= 1.0 m

Average SfM Error:

= (0.5 + 0.8 + 1.0) / 3

= 2.3 / 3

= 0.77 m

Comparison:

Stereo Vision = 0 m average error

SfM = 0.77 m average error

Conclusion:

Stereo Vision is more accurate because it has a lower average depth estimation error.

5. Pattern Recognition Accuracy Comparison

Given Data:

Traditional System: Correct Predictions = 85, Total Samples = 100

Deep Learning System: Correct Predictions = 92, Total Samples = 100

(a) Calculate Accuracy

Accuracy Formula:

Accuracy = (Correct Predictions / Total Samples) × 100

Traditional System:

= 85 / 100 × 100

= 85%

Deep Learning System:

= 92 / 100 × 100

= 92%

Comparison:

Traditional = 85%

Deep Learning = 92%

Conclusion:

Deep Learning performs better because its recognition accuracy is higher.